

AGC Take-Home — Report

Name:

Time spent:

Fill in what you got to. Write “not measured” where you did not — that is a fine answer and much better than a guess.

1. What works

Briefly: what is running, and what is not.

2. Timing

	Value
Core clock you configured	
How you confirmed it	
Worst-case block time (cycles)	
As a percentage of the 8 ms budget	

Where does the time go? Biggest consumer first.

3. Counters at the end of a full run

Counter	Value
DMA overruns	

Counter	Value
ADC overruns	
Stream drops	
Blocks processed	
Sequence gaps reported by the host script	

If any are non-zero, what happened?

4. Your AGC

	Value
How fast the gain comes down	
How fast it goes back up	
Gain limits you used	
How often you update the gain	

Anything you tried that did not work?

5. Your filter

What kind, what order, where the coefficients came from, and what it does to the signal.

6. Real-time design

Which interrupts you used and what priority you gave them, and why:

Where does data cross from interrupt context into your main loop, and why is that safe?

What happens if the serial link stops draining?

7. What you would do next

With another day:

The weakest part of what you built:

8. AI assistance

Which tools you used, and for what:

What they got wrong that you had to fix. This is the part we find most useful — be specific.

9. Anything else